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Project Description:

I am choosing to go in the direction of my Mini-AES idea. The design I create will decrypt an inputed cipher text, by using a version of AES with a reduced number of rounds. To do this, I will implement an inverse S-box, perform the reverse of the shift rows step and the mix columns step. Also, I will need to derive the correct key for each round, and implement an fsm in order to pipeline the different rounds, and reduce the amount of FFs needed. The user can press a button when the input is ready, and the led will light up when the output is ready.

One concern I have is that there will not be enough cells required to implement full AES-128 decryption. For this reason, I am considering making my design operate on a much smaller scale. As I am not yet sure how many cells I will need, I am planning on implementing my design using parameters, than scaling down accordingly. As of right now, I hope that this will mean my design can operate on at least 64 bits, but that number might become smaller later on. Another concern is how I will take in the 64 bit key and 64 bit ciphertext inputs. Since there are only 12 bits available, I was planning on having it take multiple cycles to input both the key and the ciphertext. I am not yet sure if this is the most efficient way to do this, or if there is a peripheral that could make this process easier.

A simple block diagram is shown below. The inputs to my module will be a 64 bit key and a 64 bit ciphertext, as well as a clock, reset, and button inputs. The outputs will be a ready signal (to be passed to the LED), and a 64 bit plaintext.

The only peripherals I plan on needing will be a button and an LED.